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Class 9
Chemistry
Chapter 5
NOTES

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CHEMICAL BONDING

—: (Question and Answers) :—

Question no 1:- Why do atoms react?

Answer:- Atoms other than noble gases tend to react to become stable either by losing or gaining electrons or by sharing ~~gaining~~ electrons.

Question no 2:- What is duplet or Octet rule?

Answer:- **DUPLET RULE**:- The tendency of atoms to acquire two electronic Configuration in their outermost Shell during bond formation is called "Duplet Rule".

Example:- $\text{Li} = 1s^2 2s^1$ lose 1 electron to form $\text{Li}^+ (1s^2)$

OCTET RULE:- The tendency of atoms to acquire eight electronic Configuration in their outermost Shell during bond formation is called "Octet Rule".

Example:- $\text{Na} = 1s^2 2s^2 2p^6 3s^1$ lose an electron to form $\text{Na}^+ : 1s^2 2s^2 2p^6$ & achieve noble gas Configuration.

Question no 3:- Define Chemical bond?

Answer:- **CHEMICAL BOND**:- "Some forces of attraction hold atoms together in substance. These forces are called chemical bond." These forces are electrical in nature.

These forces are electrical in nature.

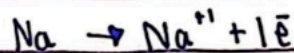
Question no 4:- What are Electropositive and electro-negative of element.

Answer:-

ELECTRO POSITIVITY:-

The tendency of metal to lose electrons is called electro positivity.

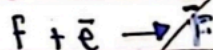
Example:-



ELECTRO NEGATIVITY:-

The tendency of Non-Metal to gain electrons is called electro negativity.

Example:-



Question no 5:- What are the two type of bonds.

Answer:- Depending on the tendency of an atom to lose or gain or share electrons, there are two types of bonds:-

- 1- Ionic bonds
- 2- Covalent bonds.

Question no 6:- Describe Ionic Bonds with the help of diagram.

Answer:- Ionic bonds are formed between two atoms, when one atom loses electron to form cation & the other atoms gain this electron to form anion.

"An ionic bond is a strong electrostatic attraction between positively charged metal ions and negatively charged non-metals ions."

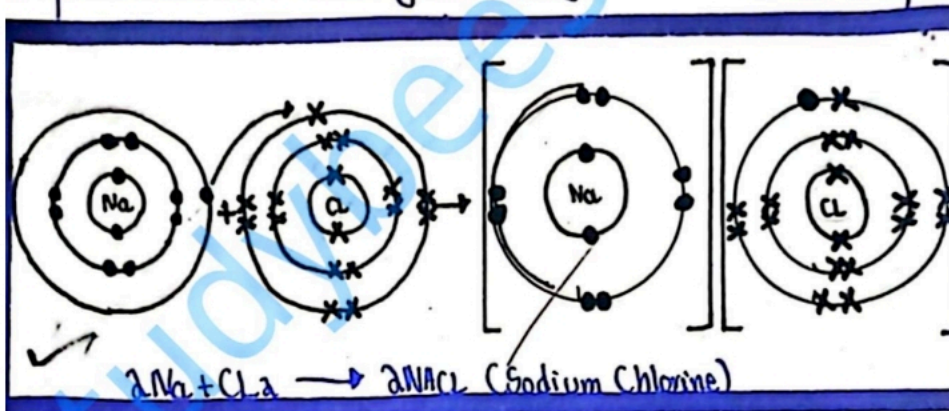
Compounds that consist of ions joined by electrostatic forces

Ionic Compound: The total positive charge of the cations must be equal to the total negative charge of the anions. This is because ionic compound are electrically neutral as a whole.

Representing ionic bond formation.

A) Na is metal and Cl is non-metal.

Metal atom tends to lose electrons & non-metal atoms tends to gain electron acquire electronic configuration of nearest noble gas. Since a Na atom has one electron in the outermost shell. It loses one electron to form Na^+ ion. Since a Cl atom has seven electrons in outermost shell. It needs one electrons to complete octet. So it gains one electron to form Cl^- ion. For every Na^+ ion, you need one Cl^- ion.



Question no 7:- How MgF_2 is formed?

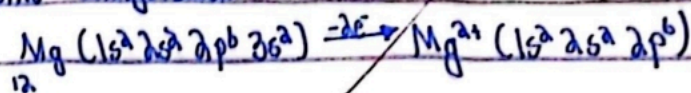
1) Mg is metal and F is non-metal.

A Mg atom has two electrons in the outermost shell. It loses 2 electrons to form Mg^{2+} ion. Since a F atom has seven electrons in the outermost shell, so it gains one electron to form F^- ion.

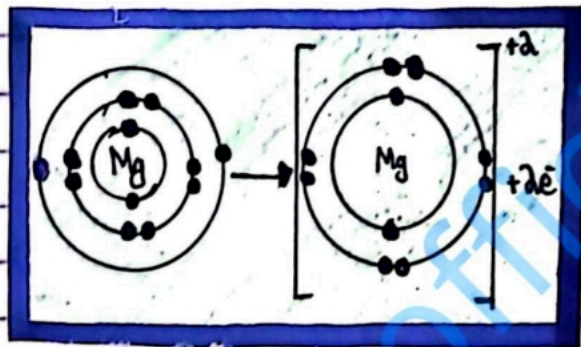
Sodium belongs to Group 1A on the Periodic table. It has only one electron in the Valence Shell. The Sodium atom loses its valence electron & is left with an octet.

Formation of Mg^{2+} ion

Electronic Configuration:



You can also represent this by electron dot structure.



Magnesium belongs to Group IIA in the periodic table. It has two valence electrons. A magnesium atom loses these electrons. A magnesium atom loses these electrons to achieve noble gas configuration.

Question no 1: Describe the formation of anion.

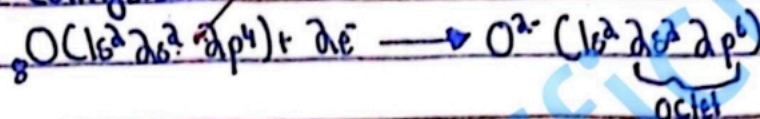
Answer:

Formation of anion by oxygen atom

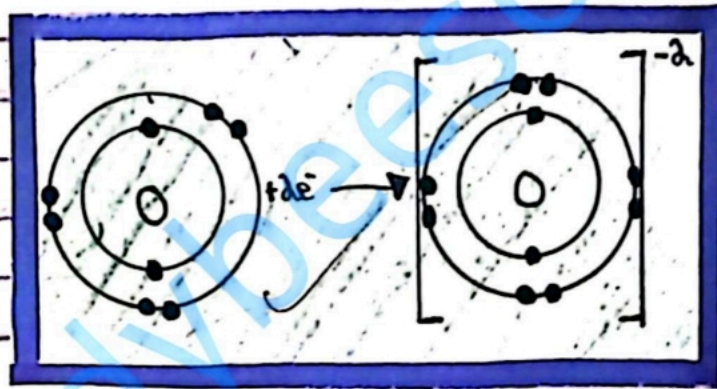
Oxygen belongs to Group VIA on the periodic table. So it has 6 electrons in its Valence Shell.

It needs two electrons to achieve noble gas Configuration.

Electronic Configuration



You can also represent this by electron dot structure,

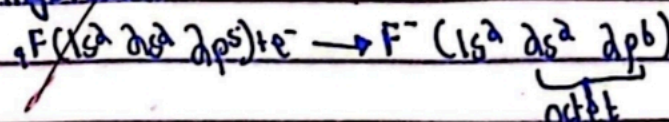


Formation of anion by Fluorine atom

Fluorine belongs to Group VIIA on the periodic table.

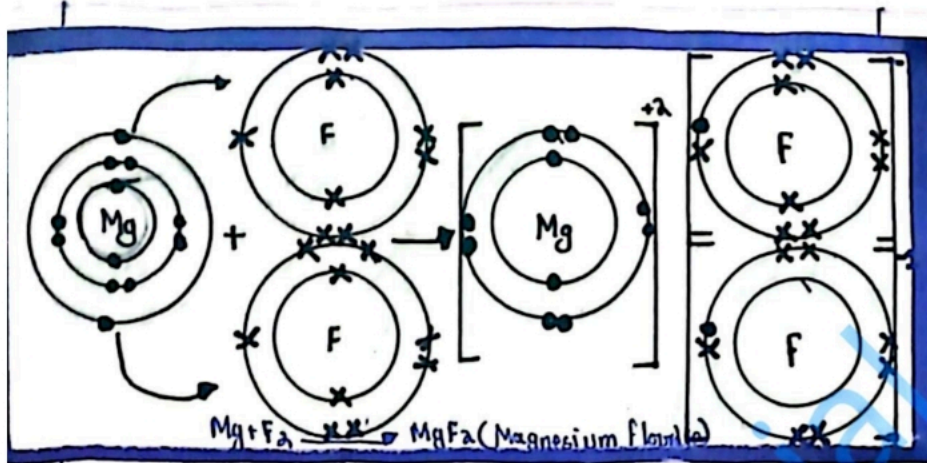
So it has seven electrons in the Valence Shell. A Fluorine atom therefore, requires only one electron to complete octet.

Electronic Configuration:



You can also represent this by electron dot structure,

has seven electrons in the outermost shell, so it gains one electron to form F^- ion.

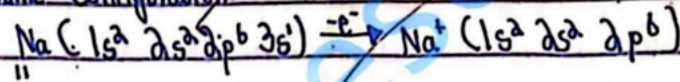


Question No 8: Describe the formation of cations.

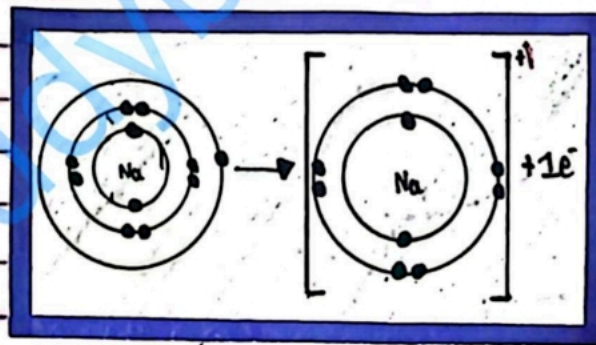
Answer.

Formation of Na⁺ ion:-

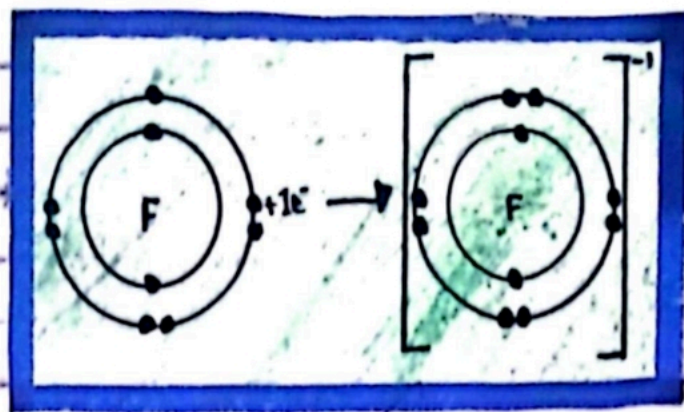
Electronic Configuration:-



You can also represent this by following electron dot structure.



Sodium belongs to Group 1A on the Periodic table. It has only one electron in the Valence shell. The Sodium atom loses its valence electron & is left with an octet.



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— :: (Concept Assessment Exercise 5.3) :: —

For each of the following pairs of atoms, use electron dot & electron cross structure to write the equation for the formation of ionic compound.

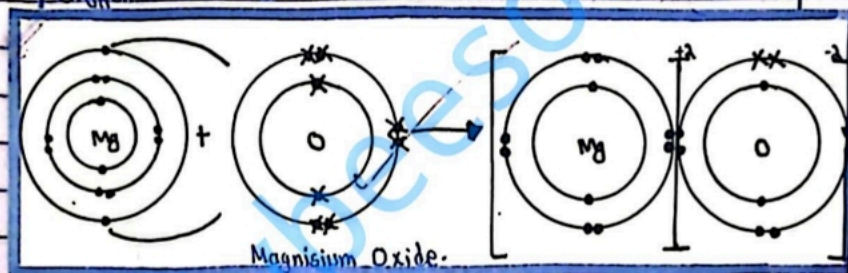
a) Mg & O

Solution:-

Mg & O

Mg atomic no = 12

Oxygen " " = 8



Mg is metal, oxygen is non-metal. A Mg has two electrons in outermost shell. So it loses two electron to form Mg^{2+} ion. Since an O atom has six electron in outermost shell, so it gains two electrons to form O^{2-} ion. In this way, they acquire nearest noble gas Configuration.

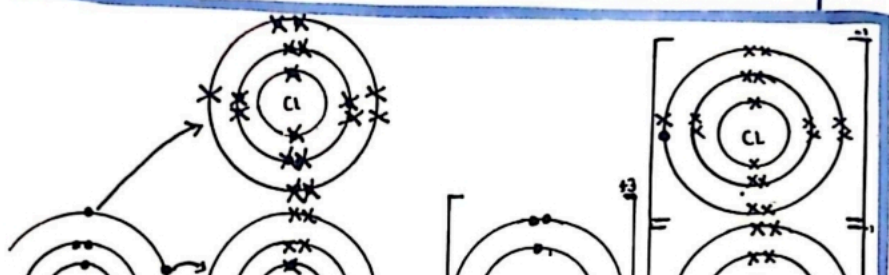
b) Al & Cl

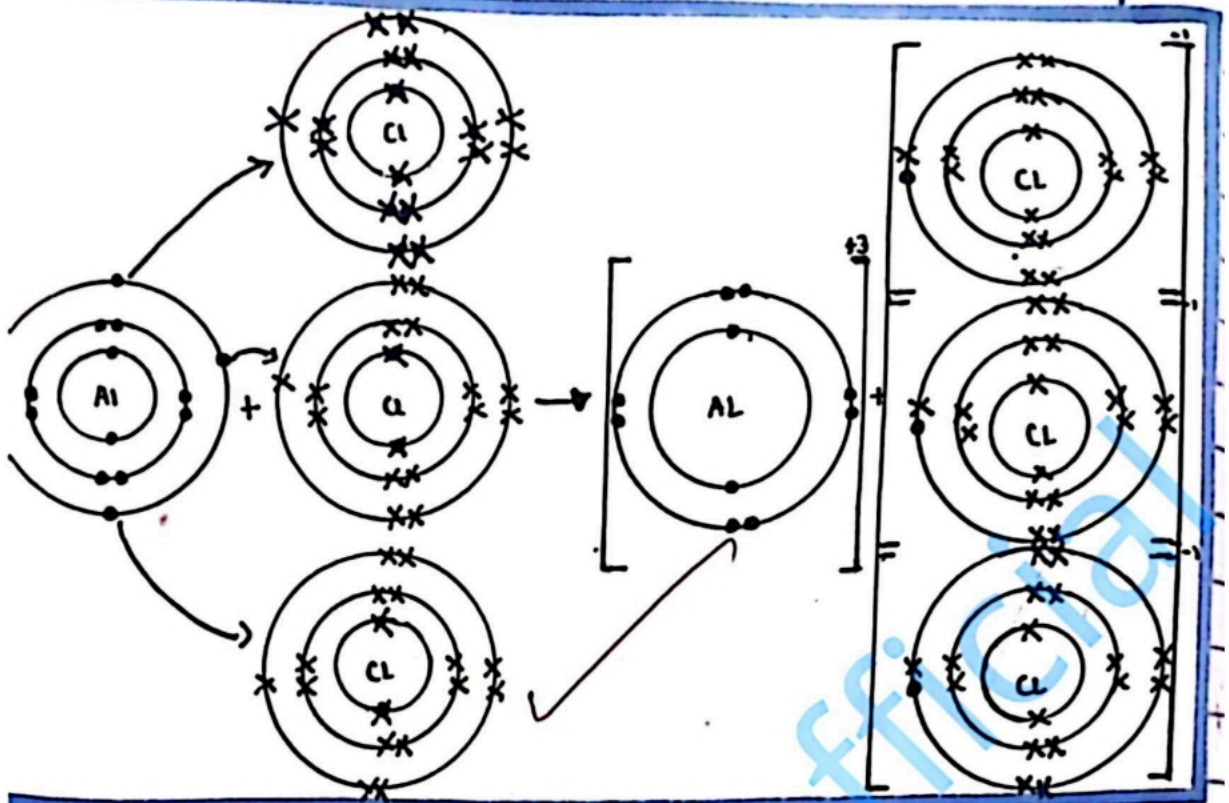
Solution:-

Al & Cl

Al atomic no = 13

Cl " " = 17





Al is metal & Cl is non-metal. Al has 3 electrons in its valence outermost shell. So it loses 3 electron to form Al^{+3} ion. Since Cl atom has 7 electrons in its outermost shell. So it gains 1 electron to form Cl^{-1} . So in this way both atoms acquire nearest noble gas configuration.

— :: (Concept Assessment Exercise 5.4) :: —

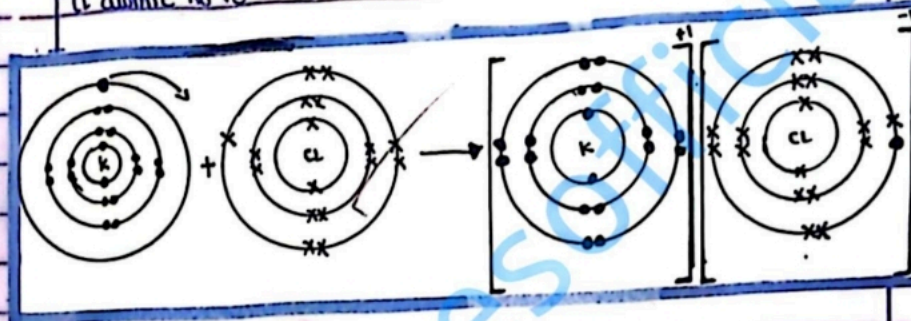
D KCL

Sol: "

KCL

"K" atomic no is = 19

"Cl" atomic no is = 17



"K" is metal it is belong to group 1A it has 1 electron in its Valence Shell. it loses 1 electron to form K^+ to achieve noble gas. Where as "Cl" is non-metal. it is belong to group VIIA it has 7 electron in its valence Shell. it gain 1 electron to form Cl^- to achieve noble gas. Together they form Potassium Chloride.

Question no 10: Define Covalent bond, explain it's types with the help of example.

Answer: COVALENT BOND:-

"A chemical bond formed by mutual sharing of electron b/w two atoms is called Covalent bond."

There are three types of Covalent Bond.

Question no 10: Define Covalent bond, explain it's types with the help of example.

Answer: COVALENT BOND:-

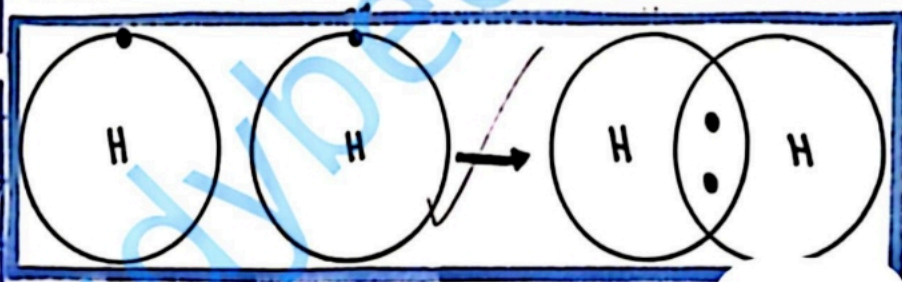
"A chemical bond formed by mutual sharing of electron b/w two atoms is called Covalent bond."

There are three types of Covalent Bond.

SINGLE COVALENT BOND:-

The bond which is formed by mutual sharing of one electron pair is called Single Covalent bond. It is represented by one straight line. (—)

Example:- Consider the formation of a Covalent bond between two hydrogen atoms. A hydrogen atom has one electron. Two hydrogen atoms share their valence electron to form a diatomic molecule.



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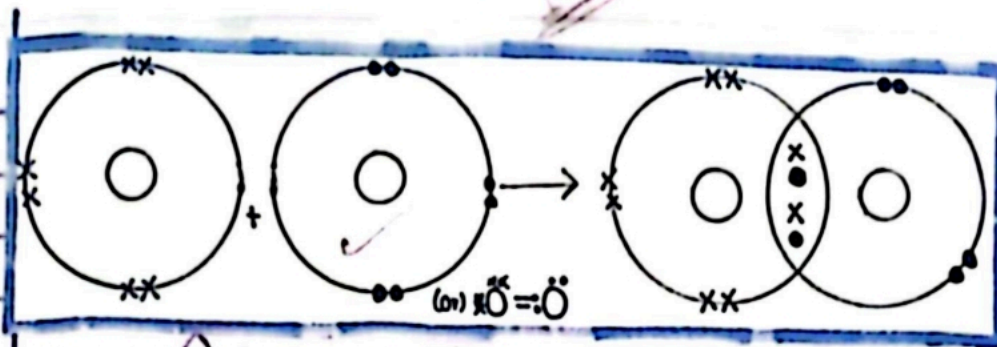
DOUBLE COVALENT BOND:-

The bond is formed by mutual sharing of two electron pairs. It is called double Covalent bond.

It is represented by two straight lines. (==)

Example:- Consider the formation of a Covalent bond between two oxygen atoms. Oxygen atom belongs to group VI-A. It has six electrons in its valence shell. It needs two electrons to complete its octet. So for sharing each O-atom contributes two electrons.



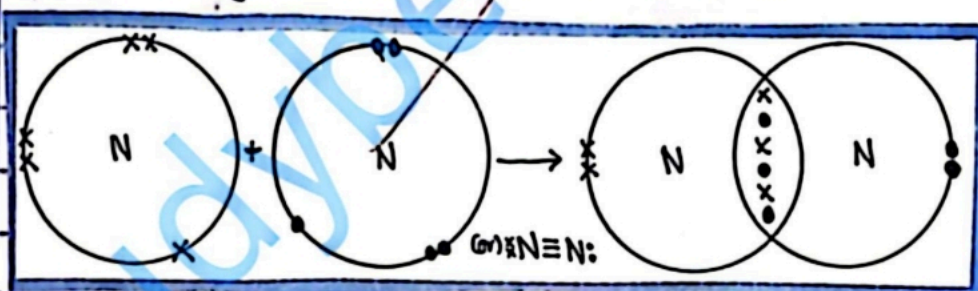


TRIPPLE COVALENT BOND:-

"The bond which is formed by mutual sharing of three electron pair is called tripple covalent bond. It is represented by three straight lines ($\equiv$)

Example:- Consider the formation of N_2 molecule.

Nitrogen atom is in group VII. It has five electron in its valence shell. It needs three electron to complete its octet. So for sharing each N-atom contributes three electrons.



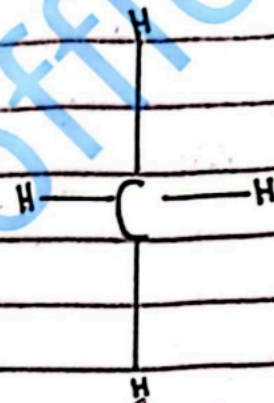
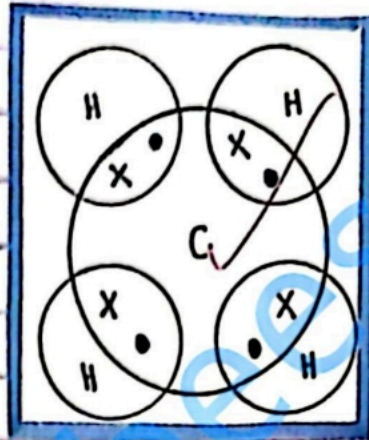
Question no 11: Draw electron cross & dot structure for the following.

Answer:-

A) CH_4

C has four electrons in the valence shell & needs four electrons to complete its octet. H has only one valence shell electron & needs one electron to complete the duplet. So C can form four single bonds with four H atoms. C is the central element.

Structure:-



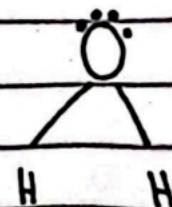
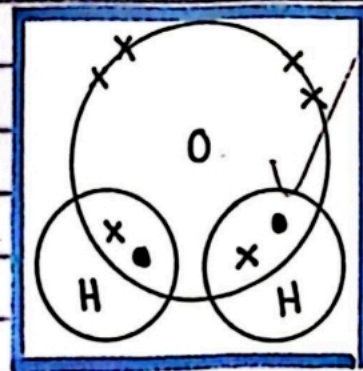
B) H_2O

O has six valence electrons $\times \ddot{\text{O}} \times$ & each H atom has one valence electron. H. So O-atom needs two electrons to complete the octet. Each H needs one electron to complete octet.

i) O is central atom

and will form two single bonds with H-atom.

Structure:-



c) CO_2

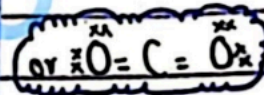
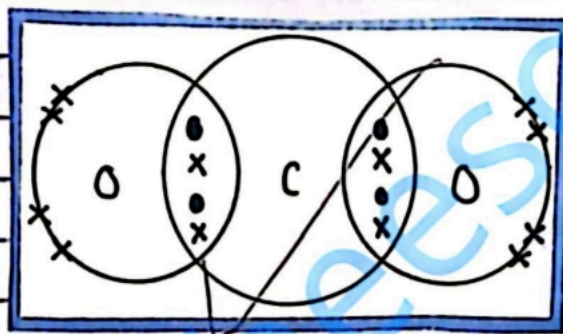
i) C has 4 electrons in the valence shell. It needs four electrons to complete octet.

ii) Each oxygen atom has 6 valence electrons & needs two electrons to have an octet.

iii) C is central atom, O is an around.

iv) Since C needs four electrons & there are only two oxygen atoms so it will share its two electrons with each O atom.

Structure:-

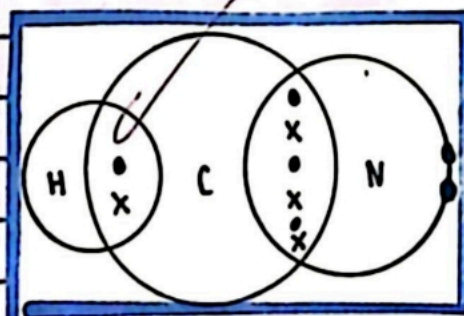


d) HCN

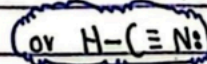
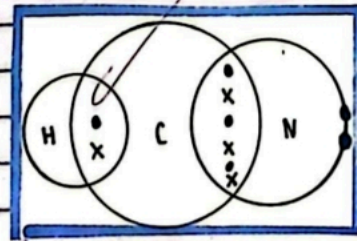
i) H has One, C has Four, N has Five electrons.

ii) C need four & N needs 3 electrons, so C share one electron with H to form a single bond & three electron with N to form a triple bond. This will satisfy octet rule.

Structure:-



Structure:



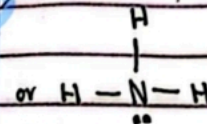
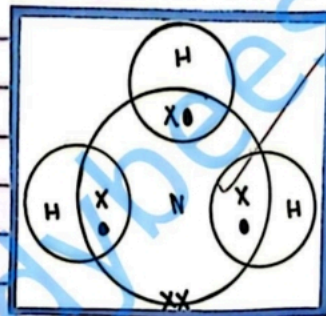
— (Concept Assessment Exercise 6.5) —

Q:- Draw electron cross & dot structure for the following molecule:-

a) NH_3

Nitrogen has 5 electrons in its valence shell & needs 3 electrons to complete its octet. Hydrogen has one electron in its valence shell & needs 1 electron to complete its duplet. So, N can form three single bond with three Hydrogen atom. Nitrogen is the central element connect the atom with the dot & cross.

Structure:-

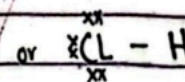
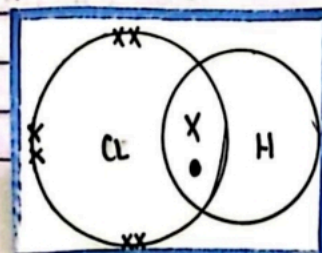


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b) HCl

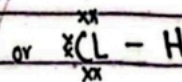
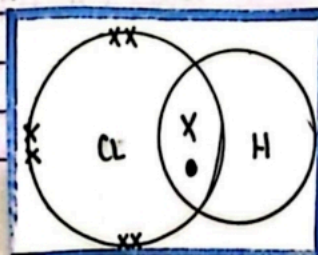
Chlorine has 7 electrons in its valence shell, it needs 1 electron to complete its octet. Whereas, H has 1 electron in its valence shell, it needs 1 electron to complete its duplet. Chlorine can form single bond with Hydrogen atom.

Structure:-



electron to complete its octet. Chlorine has 7 electrons in its valence shell, it needs 1 electron to complete its duplet. Chlorine can form single bond with Hydrogen atom.

Structure:



Question no 12:- What are Polar and non-Polar Covalent bond?

Answer:-

Polar Covalent bond:-

When two different atom share an electron pair, both atom exert different force on the shared electron pair. The more electronegative atoms pulls the shared electron pair towards itself with greater force.

Non-Polar Covalent bond:-

When two identical atom share electron pair, both atom exert in same force on shared electron pair. Such a Covalent bond is called as non-polar covalent bond, for example bond $\text{H}-\text{H}=\text{O}-\text{O}$ etc, are non covalent bond.

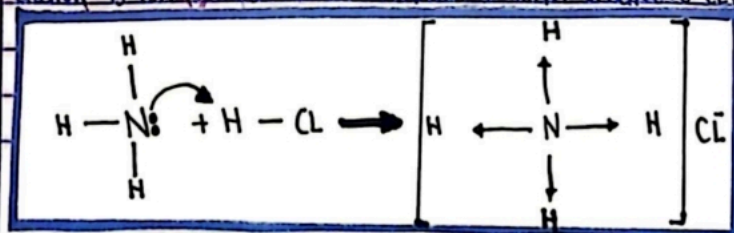
Question no 13:- Define Co-ordinate Covalent bond & explain with the help of example.

Answer:- Definition:- It is a type of covalent bond where the shared e^- pair comes from a single bond (called donor).

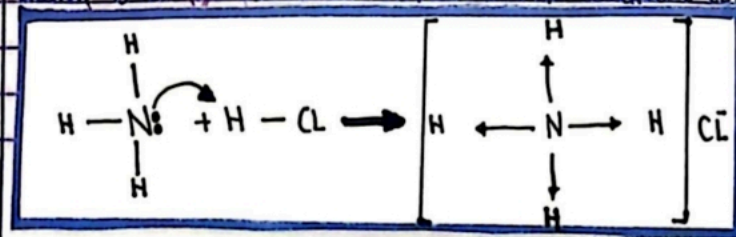
EXAMPLE:-

i) Ammonium (NH_4^+) ion:-

It is formed from the reaction of ammonia (NH_3) gas with Hydrogen chloride (HCl) gas. In NH_4^+ , the fourth hydrogen is attached by coordinate covalent bond b/c only the H nuclei is transferred from the chlorine to the nitrogen. The hydrogen's electron is left behind on the chlorine to form a negative Cl^- ion.

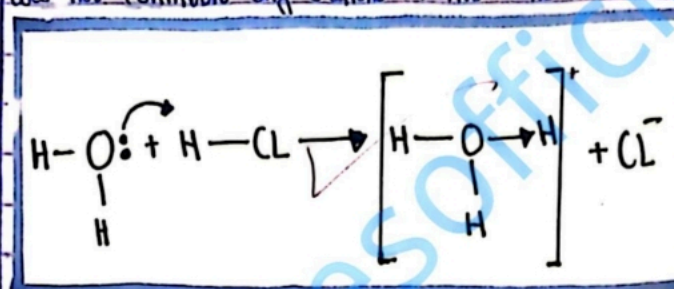


is attached by coordinate covalent bond b/c only the H nucleus is transferred from the chlorine to the nitrogen. The hydrogen's electron is left behind on the chlorine to form a negative Cl ion.



a) Hydronium ion (H_3O^+)

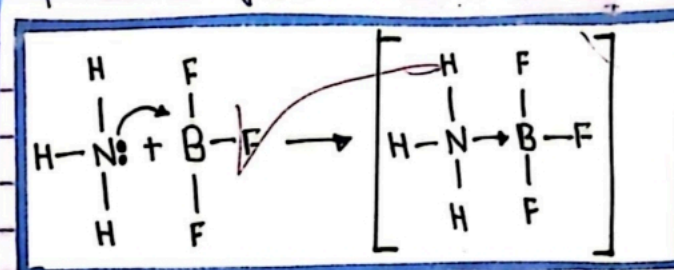
When hydrogen chloride (HCl) gas dissolves in water to make hydrochloric acid (HCl aq.) a coordinate covalent bond is formed in the hydronium ion. The hydrogen (H) nucleus is transferred to the water (H_2O) molecule, which has a lone pair of electrons to form hydronium. So H does not contribute any electrons to the bond.

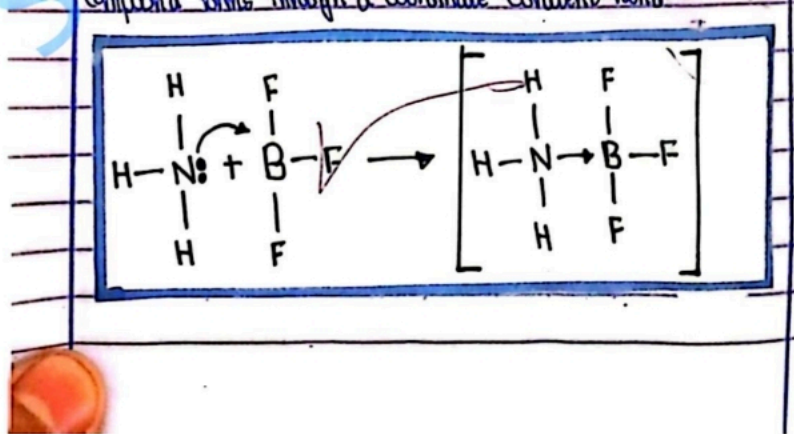


3) Ammonia Boron Trifluoride (NH_3-BF_3)

Boron trifluoride (BF_3) is a compound that does not have a noble gas structure around the Boron (B) atom. The boron only has three pairs of electrons in its valence shell & requires a pair to complete the orbital. Hence, BF_3 is electron deficient.

The lone pair on the nitrogen (N) of the ammonia (NH_3) molecule is used to overcome that deficiency, and a complex compound forms through a coordinate covalent bond.





Question No 13:- Define intermolecular forces & its types?

Answer:- Intermolecular forces:-

An intermolecular force is the attractive force that exist b/w the molecules.

Types:-

There are two types of intermolecular force

i Dipole-dipole forces.

ii Hydrogen bonding.

Question No 14:- Define dipole-dipole forces.

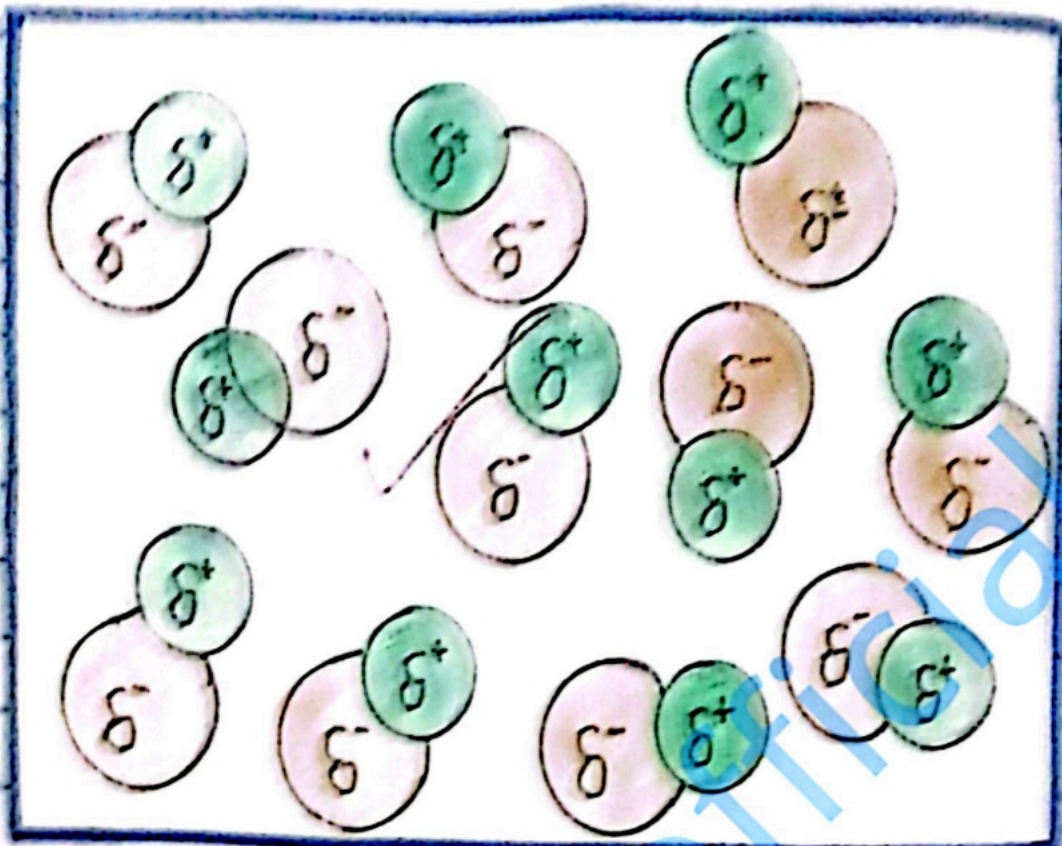
Answer:- Dipole-dipole force:-

When slightly negative end of polar molecule is weakly attracted to the slightly positive end of another molecule. Such attracting forces are called dipole-dipole interactions.

Dipole-dipole interactions occur b/w polar molecules.

Example:-

Chemically, resins are either adhesive or they form bond linkages with the material being bonded together.



Dipole -

Dipole

Interactions.

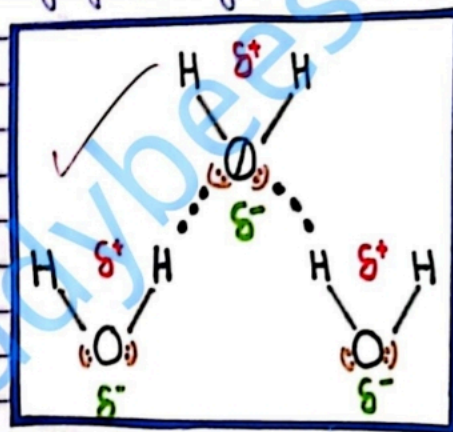


Question no 13:- Define hydrogen bonding?

Answer:- Hydrogen bonding:-

The interaction of a highly electronegative hydrogen bond with a nearby highly electronegative atom such as N , O , F is called hydrogen bond.

Molecules in which hydrogen is covalently bonded to a very electronegative atom such as O , N , F is also weakly bonded to a lone pair electrons of another electronegative atom. This other atom may occur in the same molecule or nearby molecule. This intermolecular interaction is called hydrogen bonding.



hydrogen
bonding
in
water

Question no 14:- What are the properties of intermolecular forces:-

Answer:- These intermolecular force are extremely important in determining properties of water, biological molecule. Such as proteins, DNA & Synthetic materials such as glue & resins etc. The adhesive actions of paints & dyes is developed

Question no 16:- What are the properties of intermolecular forces:-

Answer:- These intermolecular force are extremely important in determining properties of water, biological molecule. Such as proteins, DNA & Synthetic materials such as glue & resins etc. The adhesive actions of paints & dyes is developed due to hydrogen bonding. Synthetic resins bind two surface together by hydrogen bonding or dipole-dipole interactions.

Question: no 17:- What are epoxy adhesives. Write its properties & uses:-

Answer:-

Epoxy adhesives:-

~~have excellent chemical~~ Contain partially positively charged H-atom & oxygen atom containing lone pair in their molecule. Epoxy adhesive are therefore, sticky & can make H-bonds with other substance.

Properties:-

Epoxy adhesive have excellent chemical resistance, good adhesion properties, good heat resistance & they form strong & tough coating.

Uses:-

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Modern aircraft, boats & automobiles such as car, truck etc & even in space craft epoxy adhesive are ~~for~~ used for assembling, saving money & reducing ^{weight} ~~money~~. This means that glue & adhesive has become an essential item in our daily life.



assembling, saving money & reducing money. This means that glue & adhesive has become an essential item in our daily life.

Question no 18:- Write a note on nature of bonding, Structure & properties.

Answer:-

- The type of elementary particle contained: The substance can contain atoms, ions or molecule. For example, if it contains ions (such as sodium chloride) it will conduct electricity when melted or dissolved in water. In order to be soluble in water, the substance must contain ion or polar molecules.

- Particles may have ionic, covalent, metallic, or weak intermolecular force. The substance the bond, the higher the melting/boiling point and hardness of the substance.

eg:- Silicon dioxide (SiO_2) has strong covalent bond.

- The atoms in silica are difficult to separate, making it very hard & difficult to melt.

- On the other hand carbon dioxide has strong covalent bond. b/w the C & O atoms. but these molecules have weak intermolecular force b/w them. The molecules are therefore easily separated & so CO_2 has a low melting/boiling point.

- Particles may be arranged in planes (for example, polymers) in layers (for example, clays, graphite) or in a variety of 3-dimensional network. In graphite atoms are arranged in 2-dimensional layers.

- Diamond has a large three-dimensional network of carbon atoms, which make it the hardest substance on earth. Met 106/144
point structure. metallic bonding is strong, most metals have very high melting/boiling point & are thermally stable.

Question no 19:- Write a note on the conduction of electricity in ionic compounds.

Answer:- When an ionic compound is melted or dissolved in water, it is ionized. Its ions move freely in molten or aqueous

Question no 19:- Write a note on the Conduction of electricity in ionic Compounds.

Answer:- When an ionic Compound is melted or dissolved in Water. It is ionized, its ions move freely in molten or aqueous Solution. Therefore electricity can pass through a molten ionic Compounds or its aqueous Solution.

Question no 20:- Write a note on Conduction of electricity through acid?

Answer:- Covalent Compounds have no free charged particles, So they do not Conduct electricity. However, some Covalent Compounds Conduct electricity when dissolved in water. For instance, acids like HCl , H_2SO_4 , HNO_3 etc. When these acid are dissolved in water, they ionize & form high concentrations of H^+ ions & negatively charged ions. These ions can move easily in aqueous Solution. Therefore, aqueous Solutions of acids Conduct electricity.

Question no 21:- What are the properties of Covalent Compound.

Answer:-

- The intermolecular forces b/w their molecules are weaker than the covalent bonds.
- Covalent Compounds have low melting & boiling point.
- They are the poor conductor of ~~heat~~ electricity.

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weaker than the covalent bonds.

- Covalent Compounds have low melting & boiling point.
- They are the poor conductor of ~~the~~ electricity.

Question no 22:- Why covalent compounds usually have much lower melting ^{point} than ionic compounds. Explain?

Answer:- Covalent compounds usually have much lower melting point than ionic compounds because ionic compounds involve breaking the ionic bond. Breaking the electrostatic force b/w ions requires large amount of energy. Ionic compounds have high melting/boiling point.

Question no 23:- Define graphite, structure & the uses of graphite.

Answer:-

Graphite:-

Graphite name is derived from the Greek word "graphein" meaning "to write". It is commonly called black lead.

Graphite is allotrope of carbon.

Structure of Graphite:-

In graphite, each carbon atom is linked with 3 other carbon atoms by a single covalent bond resulting in the hexagonal rings arranged in a layer. It has 2-dimensional layer structure.

Uses of graphite:-

- Graphite is used in many refractory items such as carbon refractory bricks.
- It is used as an anode in electrolytic process.
- It is used in the manufacturing of carbon brushes and electric motors.

- Graphite material is used in engineering sectors in the making of thrust & journal bearing, piston rings & valves.

- Other applications of graphite including metallurgy, as lubricants, & in the production of paints & pencil.

Question no 24: Define diamond, structure, properties and the use of diamond.

Answer:

Diamond:-

Diamond is an allotrope of carbon in which the carbon atoms are arranged in a diamond cubic crystal lattice.

Structure of Diamond:-

In a diamond, the carbon atoms are arranged tetrahedrally. Each carbon atom is attached to other four carbon atoms 1.544×10^{-10} meter away with a C-C-C bond angle of 109.5° . It is a strong rigid 3-dimensional structure that results in an infinite network of atoms.

Properties & use of diamond:-

The giant structure & extensive covalent bonding in diamond renders it extraordinary hardness, elasticity, high yield strength, less conductivity & chemical inertness. The properties of diamond are below:-

- Diamonds are most commonly used in ornaments like rings, necklace, earring etc.

- Its property of hardness used to drill, grind or cut material. Hence, some blades used for cutting & drills in the industry.

used diamond.

• Diamonds are used in making of medicine & beauty product. They are also used in making of medical tools like tools used in Cataract Surgery.

• Diamond produce high-quality sound because they are hard & vibrate easily at high speed. It is also used in all equipment & high quality recorders.

Question No 25:- Differentiate b/w ionic & Covalent Compounds?

Answers-

Ionic Compounds	Covalent Compounds
• Ionic Compounds are strong but they are brittle.	• Covalent Compounds with joint structure, such as diamond, silicon etc.
• Ionic Compounds are poor conductors of electricity.	• They are very strong and hard.
• When they melted or dissolved in water, they conduct electricity.	• diamond is used in cutting & drilling tools.
• They have high melting point.	• used to make ceramics, glass & refractories.
	• poor conductor of electricity.

Question No 26:- Define metallic bonds & properties.

Answer:- In metals, the valence electrons are not confined to individual atoms. These electrons are called free electron. Metals atoms lose these electrons & form positive ions. The free electrons can move throughout the entire metal structure, forming electron sea.

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Answer:- In metals, the valence electrons are not confined to individual atoms. These electrons are called free electrons. Metals atoms lose these electrons & form positive ions. The free electrons can move throughout the entire metal structure, forming electron sea.

The metal cations are held together by the strong electrostatic attractive force b/w the metal cations & negatively charged electron sea. This force gives metals their unique properties.

This type of bonding is known as metallic bonding.

The properties of metals that are a consequence of metallic bonding include:-

- Malleability.
- Ductility.
- High melting & boiling point.
- High electrical & thermal conductivity metallic lustre.

Question no 27:- Describe the structure & properties of Metals which make it suitable for industrial purposes.

Answer:-

- Metals have giant structures.
- They have very high melting & boiling point.
- They are malleable & ductile.
- They can be drawn into wires & sheets.
- Metals are good conductor of heat & electricity.